

SPARC-cyclix

User guide

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[Contributors](#)

[Citation](#)

[Acknowledgements](#)

Contributors

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Citation

If you publish work using/regarding SPARC-cyclix, please cite the following article, in addition to SPARC citations:

- <https://doi.org/10.1103/PhysRevB.103.035101>,
<https://doi.org/10.1103/PhysRevB.100.125143> (initial
developments, extended systems),
<https://doi.org/10.1016/j.jmps.2016.08.007> (initial
developments, isolated systems)

Acknowledgements

- U.S. National Science Foundation (NSF): 1553212

Input file options

Input file options for SPARC-cyclix, in addition to SPARC:

Cyclix

TWIST_ANGLE | BC | CELL | COORD | COORD_FRAC |
EXCHANGE_CORRELATION | KPOINT_GRID |

TWIST_ANGLE

Type

Double

Unit

rad/Bohr

Default

0

Example

TWIST_ANGLE: 0.0045

Description

External twist per unit length applied on the nanotube.

Remark

If using helical symmetry (D C H), we also have to add the intrinsic twist.

Type

Character

Unit

No Unit

Default

None

Example

BC: D C P

Description

A set of three whitespace delimited characters specifying the boundary conditions in the lattice vector directions, respectively. P represents periodic boundary conditions, D represents Dirichlet boundary conditions, C represents cyclic boundary conditions and H represents helical boundary conditions.

Remark

BC: D C P and BC: D C H are the cyclix cases where the former uses only cyclic boundary condition and latter uses both cyclic and helical boundary conditions.

CELL

Type

Double

Unit

Bohr

Default

None

Example

CELL: 30.3498 0.2991 3.1359

Description

A set of three whitespace delimited values specifying the cell lengths in the radial, angular and periodic/helical directions, respectively.

Remark

Angular direction length is equal to $2\pi/\Gamma$, where Γ is the group order in the cyclic direction.

COORD

Type

Double

Unit

Bohr

Default

None

Example

COORD:

0 0 0

2.5 2.5 2.5

Description

The Cartesian coordinates of atoms of a `ATOM_TYPE` specified before this variable. If the coordinates are outside the fundamental domain (see [CELL](#)) in the periodic and cyclic/helical directions (see [BC](#)), it will be automatically mapped back to the domain.

Remark

For a system with different types of atoms, one has to specify the coordinates for every `ATOM_TYPE`. One can also specify the coordinates of the atoms using `COORD_FRAC`.

COORD_FRAC

Type

Double

Unit

None

Default

None

Example

COORD_FRAC:

0 0 0

2.1 0.5 0.5

Description

The fractional coordinates of atoms of a `ATOM_TYPE` specified before this variable. $\text{COORD_FRAC}(i, j) \times \text{CELL}(j)$, ($j = 1, 2, 3$) gives the coordinate of the i^{th} atom along the j^{th} direction. $j = 1$ is along radial, $j = 2$ is along angular and $j = 3$ is along periodic/helical direction.

Remark

For a system with different types of atoms, one has to specify the coordinates for every `ATOM_TYPE`. One can also specify the coordinates of the atoms using `COORD`.

EXCHANGE_CORRELATION

Type

String

Unit

No unit

Default

No Default

Example

```
EXCHANGE_CORRELATION:  
LDA_PW
```

Description

Choice of exchange-correlation functional. Options are LDA_PW (Perdew-Wang LDA), LDA_PZ (Purdew-Zunger LDA), GGA_PBE (PBE GGA), and the finite-temperature KSDT (LDA), corrKSDT (LDA, SPIN_TYP = 0 only) and KDT16 (GGA, SPIN_TYP = 0 only)

Remark

For spin-polarized calculation ($\text{SPIN_TYP} = 1$), LDA_PW (Perdew-Wang LDA) and GGA_PBE (PBE GGA) are available.

KPOINT_GRID

Type

Integer array

Unit

No unit

Default

1 1 1

Example

KPOINT_GRID: 1 5 10

Description

Number of k-points in each direction of the Monkhorst-Pack grid for Brillouin zone integration. In SPARC-cyclix, we don't have any k-points along the radial direction as it is vacuum. The angular k-point is referred as ν point and the periodic/helical k-point is called η point.

Remark

ν point should be an integer factor of the group order Γ .

FAQs

Q: How can SCF convergence be made better in SPARC-cyclix calculations?

A: In cases of smaller radius and finer mesh, we need to use higher Chebyshev Degree. Increasing the option `CHEB_DEGREE` should help with the SCF convergence.

Q: Are pseudopotentials with non-linear core corrections and relativistic effect supported by SPARC-cyclix?

A: Yes.

Q: Is non-collinear spin calculation supported by SPARC-cyclix?

A: No, currently we only have the collinear spin calculations available in SPARC-cyclix.